Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/18/2004
PI Name:	Stone, Jennifer Ph.D.		
Project Title:	The role of the homeobox transcription factor cProx1 in the development of the vestibular periphery in chickens		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2000 Space Biology 00-OBPR-01
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Contact Monitor:		**Contact Phone:**	
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	The vestibular system coordinates motor responses to changes in head position and movement and is adaptable to different gravitational conditions. These features make vestibular function critical for space travel. Vestibular function depends upon peripheral receptors (hair cells) located in five inner ear organs and the nearby VIIIth cranial ganglion neurons that connect hair cells to the brain. These cells form during embryogenesis from the otocyst. There is evidence that development of the vestibular periphery is affected by space travel. Elucidation of the factors mediating microgravity-induced changes in vestibular development requires an understanding of normal inner ear formation. However, many aspects of normal otocyst development, including how the sensorineural regions become specified and acquire their normal cellular organization, remain unclear. The proposed experiments will examine the role of the homeodomain transcription factor, cProx1, in development of the		

Task Description:	vestibular epithelium and ganglion in normal chickens. This factor, which is critical for development of the lens, lymphatic system, and liver in mice, is highly expressed in the chicken otocyst in regions that will form the vestibular epithelia and ganglia. cProx1 expression is down-regulated after hair cells have formed, and it is re-expressed in mature chickens during hair cell regeneration. We will characterize temporospatial patterns of cProx1 mRNA and protein expression in the developing otocyst using in situ hybridization and immunofluorescence. In addition, cProx1 gene expression will be perturbed in ovo to test its role in sensory specification of the otocyst and in cell proliferation, differentiation, and patterning within sensory patches. Finally, the role of the hindbrain and periotic mesoderm in regulating the localized expression of cProx1 will be examined by transplanting otocysts into the hindbrain and by rotating otocysts in situ relative to the hindbrain. This research aims to expand upon our growing understanding of the molecular interactions necessary for vestibular end organ development in vertebrates.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The peripheral vestibular system is responsible for sensing head position and movement and is therefore critical for comfort and survival on Earth as well as in space. The goal of my research is to understand how the vestibular end organs and neurons are generated during development and regeneration in the chicken. This series of studies investigates the role of a homeobox transcription factor, cProx1, in the specification of sensory and neural cells within the developing inner ear. Identification of regulatory molecules that are necessary for otocyst morphogenesis and cellular patterning will lead to a better understanding of normal development and will help us unravel the cellular mechanisms underlying vestibular dysfunction.
Task Progress:	This year, our efforts were aimed at determining the function of the transcription factor cProx1 in the avian otocyst by misexpression and in examining the effect of otocyst rotation on the spatiotemporal expression patterns of cProx1 in the avian otic epithelium. We have been attempting to misexpress cProx1 by electroporating the pMES-Prox plasmid, which encodes full-length cProx1 and GFP under a CMV/B-actin promoter/enhancer system and an IRES. pMES encoding GFP has served as a negative control for these experiments. Using the pMES/Prox plasmid, we have achieved high levels of concurrent cProx1 and GFP expression in targeted regions of the otic epithelium of embryonic chickens. The overexpressed cProx1 becomes localized to the nucleus, where it can have its function. Electroporated ears appear to normally develop gross morphology. We are now poised to conduct the systematic experiments with these constructs to determine if the normal course of cellular differentiation is altered in cells that misexpress cProx1. We have also performed in ovo otocyst rotations. The main challenge with this method has been improving surgical accuracy to attain higher survival rates of the embryos. The main complication is embryo death due to internal bleeding. However, we now achieve a better survival rate, and the operated ear appears to develop normally, as demonstrated by microscopic analysis of cryostat sections. We are now ready to complete the study by assessing if the pattern of cProx1 expression is altered in rotated ears.
Bibliography Type:	Description: (Last Updated: 09/19/2006)
Articles in Peer-reviewed Journals	Stone, JS; Shang, JL; Tomarev, S. "cProx1 immunoreactivity distinguishes progenitor cells and predicts hair cell fate during avian hair cell regeneration." Developmental Dynamics 230(4):597-614, Aug 2004. , Aug-2004